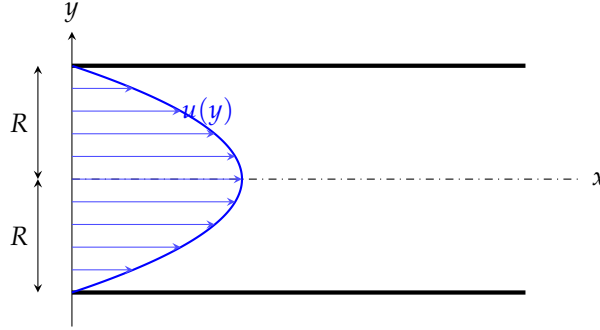


1 | Simulating the Poiseuille Flow

Poiseuille's Flow describes a laminar steady flow of an incompressible **viscous** fluid through a cylindrical pipe with a constant cross-section.

1.1 The Parabolic Velocity Profile



We use cylinder coordinate and set

$$\mathbf{u} = (u_r, u_\theta, u_z) = (u_r, u_\theta, w) = (u_r, 0, w(r))$$

Ideally, the velocity is zero at the wall, i.e. $w(R) = 0$. From the **Mass Conservation**

$$\nabla \cdot \mathbf{u} = w_z + \frac{1}{r}(ru_r)_r + \frac{1}{r}u_\theta = 0 \quad (1.1)$$

Since w is constant in z direction, then $w_z = 0$, so we have $u_r = 0$. Now by newtonian law in r direction.

$$\frac{p_r}{\rho} = 0 \quad \Rightarrow \quad p = p(z)$$

The pressure only depends on z direction. Now since

$$\Delta w = \frac{1}{r}(rw_r)_r$$

Then the z component of the N-S equation is

$$\frac{P_z}{\rho} = \mu \frac{1}{r}(rw_r)_r$$

Then using the boundary condition $w(R) = 0$, we have

$$w(r) = \frac{P_z}{4\mu}(r^2 - R^2) \quad (1.2)$$

This is the velocity profile for the Poiseuille Flow.

■ 1.1.1 Reynolds Number

The existence of the parabolic velocity profile is based on the assumption that the flow is viscous. The Reynolds number should be small in order to neglect the inertial term.

$$Re = \frac{\rho U D}{\mu} < 2000 \quad (1.3)$$

For a standard case, let's compute the Reynolds number explicitly. Let's look at a biological example: how much a small amount of plaque buildup affects blood flow in an artery. The constants :

1. Length of artery section (L): 0.05 meters
2. Pressure Drop (ΔP): 400 Pascals (the force from the heart)
3. Viscosity of blood (η): 0.004 Pa·s
 - (a) Healthy Artery (R_1): Has a radius of 0.002 meters (2 mm).
 - (b) Narrowed Artery (R_2): Plaque has reduced the radius by just 20%, making it 0.0016 meters (1.6 mm).

■ 1.1.2 Total Flux

The average velocity is given by

$$\bar{w} = \frac{1}{\pi R^2} \int_0^R 2\pi r \frac{P_z}{4\nu} (R^2 - r^2) dr = \frac{P_z R^2}{8\nu}$$

Now since we are adding a constant force F on the right boundary of the tube, then

$$P_z = \frac{F}{\pi R^2} \cdot \frac{1}{L}$$

Where L is the length of the tube. Then we have

$$\bar{w} = \frac{F}{8\pi L \nu}$$

Formula to the Reynolds number for this system is approximatedly

$$Re = \frac{\rho \bar{w} R}{\nu} = \frac{\rho F}{8\pi L \nu^2}$$

1.2 2D Poiseuille's Flow

Now imagine we have a 2D channel with length L and radius R . What is the velocity profile in this case? Starting from mass conservation.

$$\nabla \cdot \mathbf{u} = u_x + v_y = 0 \quad \Rightarrow \quad u_x = 0$$

The Newtonian law in x direction is

$$\frac{p_x}{\rho} = \nu \Delta u = \nu u_{yy}$$

Since $u_x = 0$, integrate the equation and use the boundary condition $u(-R) = u(R) = 0$, we have

$$u(y) = \frac{p_x}{2\nu}(y^2 - R^2) \tag{1.4}$$

Since p decreases so $p_x < 0$, then the sign is positive.